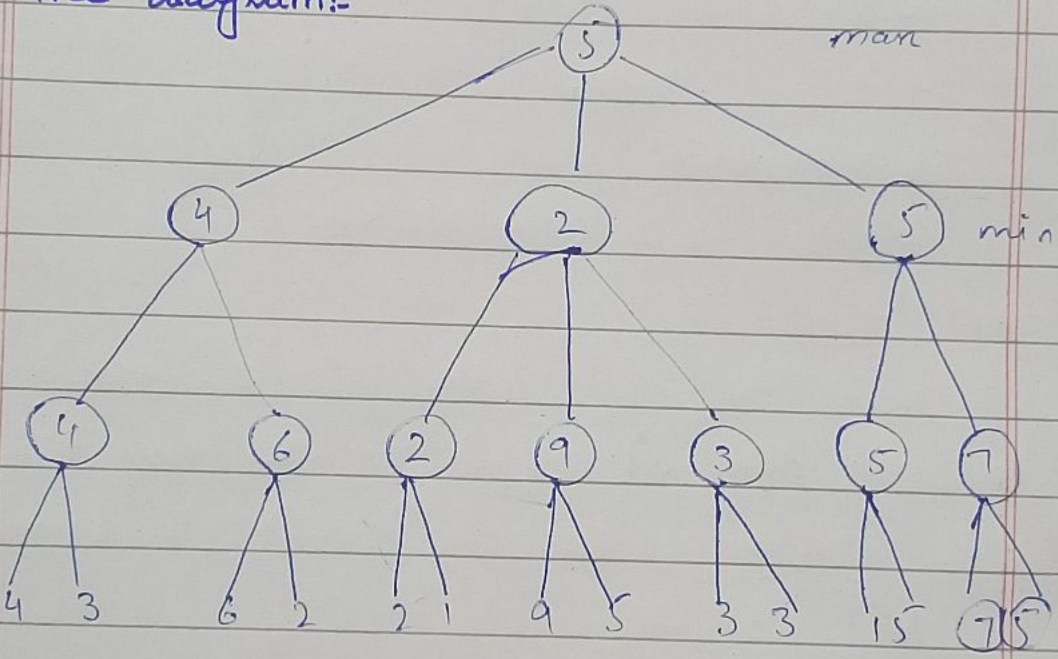
LAB # 05

Question # 01 Minimax Algorithm

Objective :-

Find the final value of the root node using the Minimax algorithm

Tree diagram:-



Max nodes (Bottom to top)

Max slects highest value

values		Max	values		Max
4, 3	→	4	3, 3	→	3
6,2	→	6	1,5	→	5
2, 1	→	2	7,5	→	7
9, 5	→	9			

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MIN nodes :-

MIN selects the lowest values

A = (6, 4) ⇒ 4

B = (2, 9, 3) ⇒ 2

C = (5, 7) = 5

Root Max :-

Max (4, 2, 5) ⇒ 5

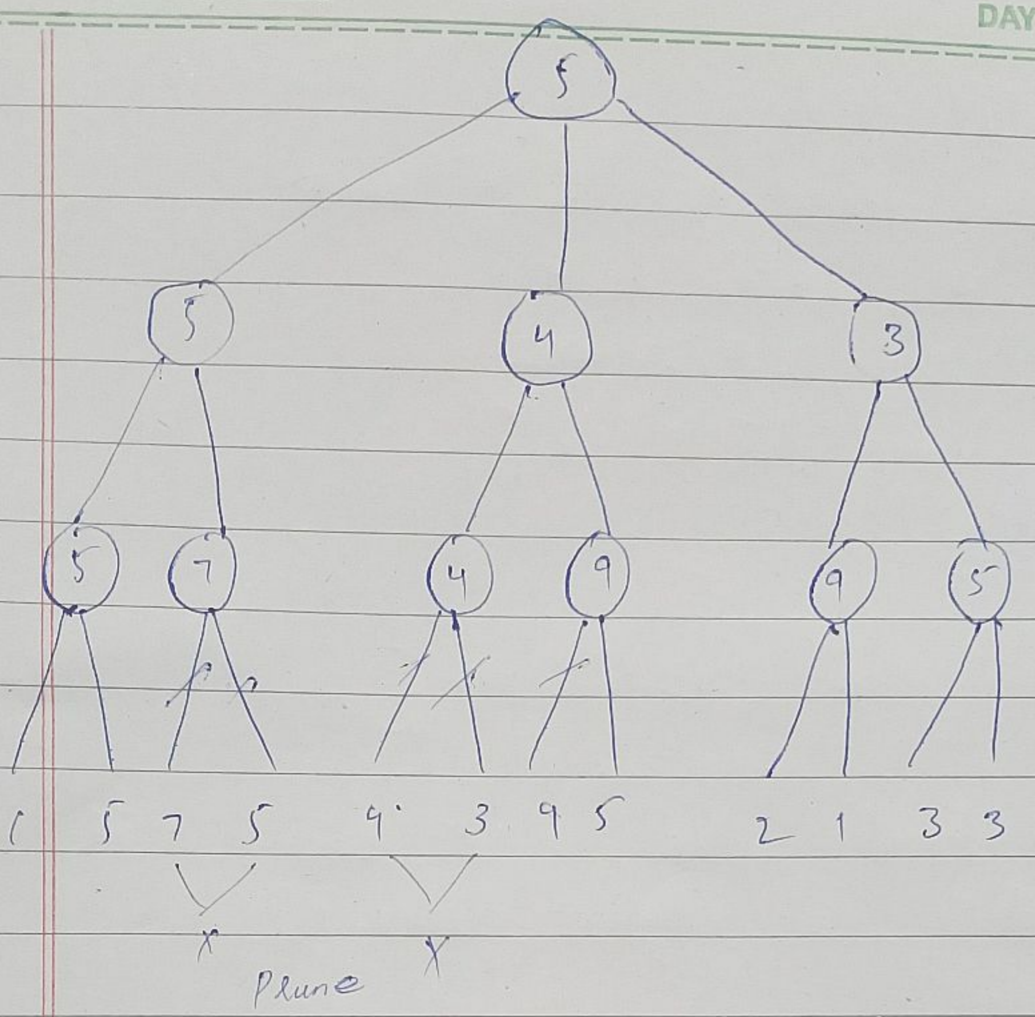
Final value is 5 because Max selects the highest value at the Root.

Question # 03

Alpha - Beta Pruning

Find the final Miniman value while removing unnecessary brancehes using Alpha-Beta pruning.

For better pruning the visiting order is changed to C, A, B



calculate node C:

$$C = \text{Min}(\text{Max}(1,5), \text{Max}(7,5))$$

$$C = \text{Min}(5,7) = 5$$

calculate node A:

First child

$$\text{Max}(4,3) = 4$$

Now

$$\alpha = 5 \quad , \quad \beta = 4$$

since $\alpha \geq \beta$

So $A = 4$

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Calculate Node B

First child

$$\text{Max }(9,5) = 9$$

second child

$$\text{Max }(3,3) = 3$$

Now the min node gets 3. Since 3 is less than or equal to the root's Alpha (5) the remaining branch (2,1) is pruned

So $B = 3$

Final Root value

$$\text{Max }(5,4,3) = 5$$